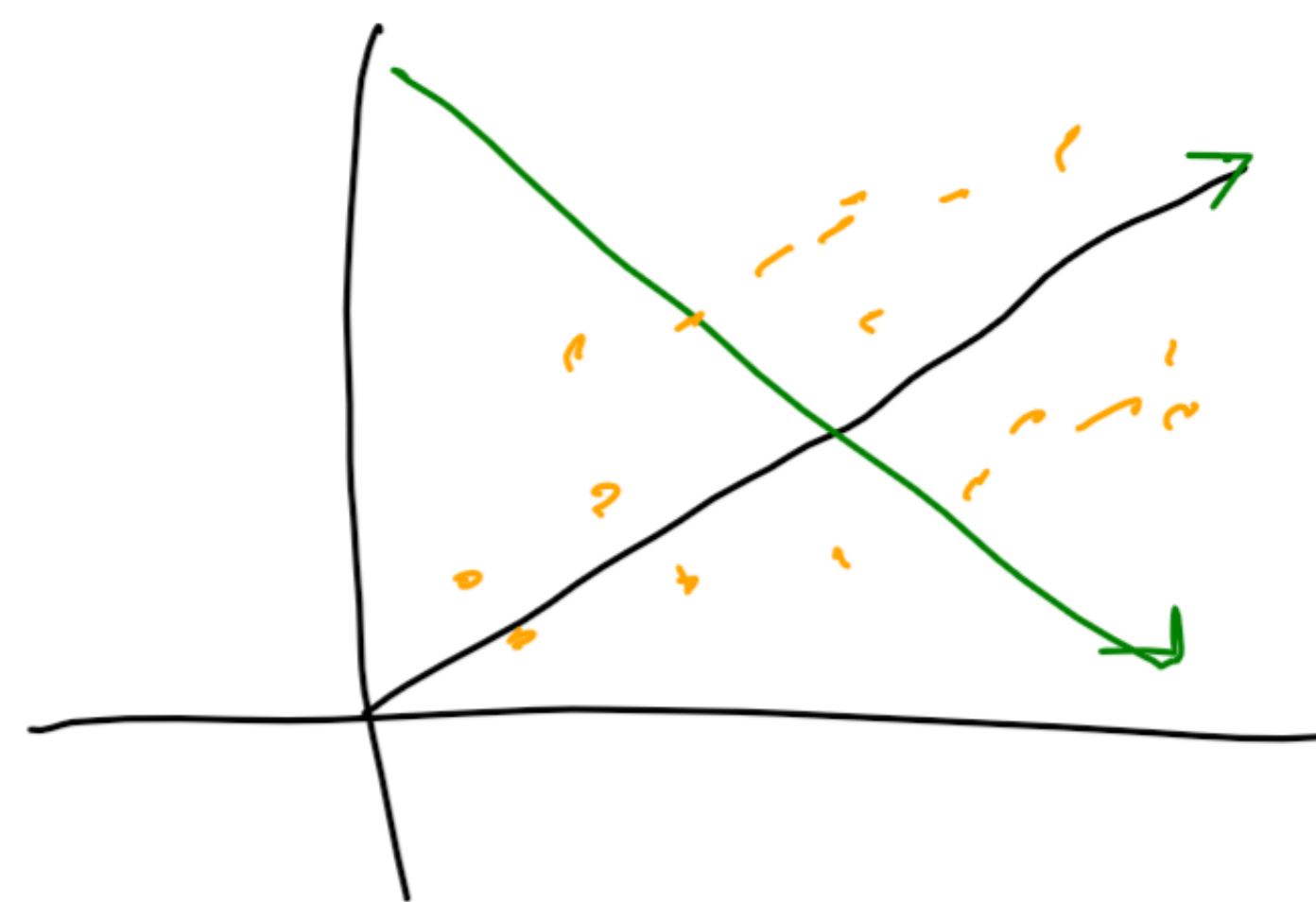


SUPERVISED LEARNING

- Regression
 - Linear Reg. ($y \sim x$)
 - Multiple Linear Reg ($y \sim x_1 + x_2 + x_3$)
- Classification



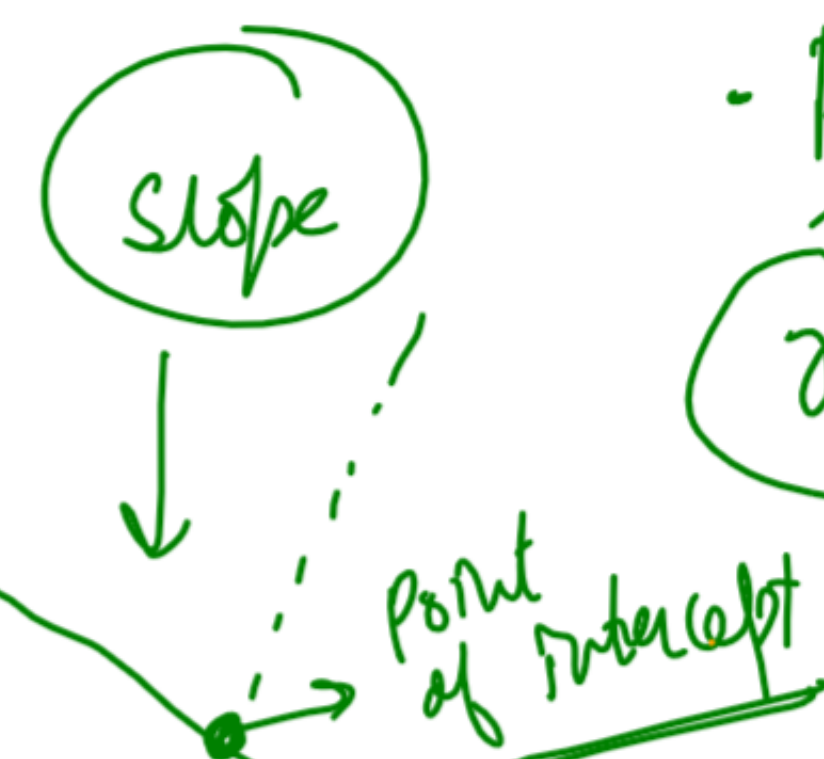
Simple Linear Regression

mathematical Relationship

$$\hat{y} = \beta_0 + \beta_1 x$$

intercept

Slope



Correlation

Directional Relationship

$$r = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sqrt{\sum (x - \bar{x})^2 \sum (y - \bar{y})^2}}$$

Mathematical function

IV	DV
5	8
6	9
3	2
4	3

n=4

x y

$$\beta_0 = b_0 = \frac{1}{n} (\sum y_i - b_1 \sum x_i) = \frac{1}{4} (22 - (2.6 \times 18)) = -6.2$$

$$\beta_1 = b_1 = \frac{\sum x_i y_i - (\sum x_i)(\sum y_i)/n}{\sum x_i^2 - (\sum x_i)^2/n} = \frac{112 - (18)(22)/4}{86 - (324)/4} = \frac{13}{5} = 2.6$$

avail	Calculated	↓
x_i	y_i	$x_i \cdot y_i$
5	8	5 × 8 = 40
6	9	6 × 9 = 54
3	2	3 × 2 = 6
4	3	4 × 3 = 12
$\sum x_i = 18$	$\sum y_i = 22$	$\sum x_i y_i = 112$
$\sum x_i^2 = 86$		

$$\sum (x - \bar{x})(y - \bar{y})$$

test value

minimize

$$\text{error} = \hat{y} - y$$

$$e = (\hat{y} - y)^2$$

$$\hat{y} - y = e$$

6.8 - 8 = -1.2	$(-1.2)^2 = 0.16$
9.4 - 9 = 0.4	$(0.4)^2 = 0.16$
1.6 - 2 = -0.4	$(-0.4)^2 = 0.16$
4.2 - 3 = 1.2	$(1.2)^2 = 0.16$

$$\sum (y - \hat{y})^2 = 0$$

$$\hat{y} = \beta_0 + \beta_1 x - e_i$$

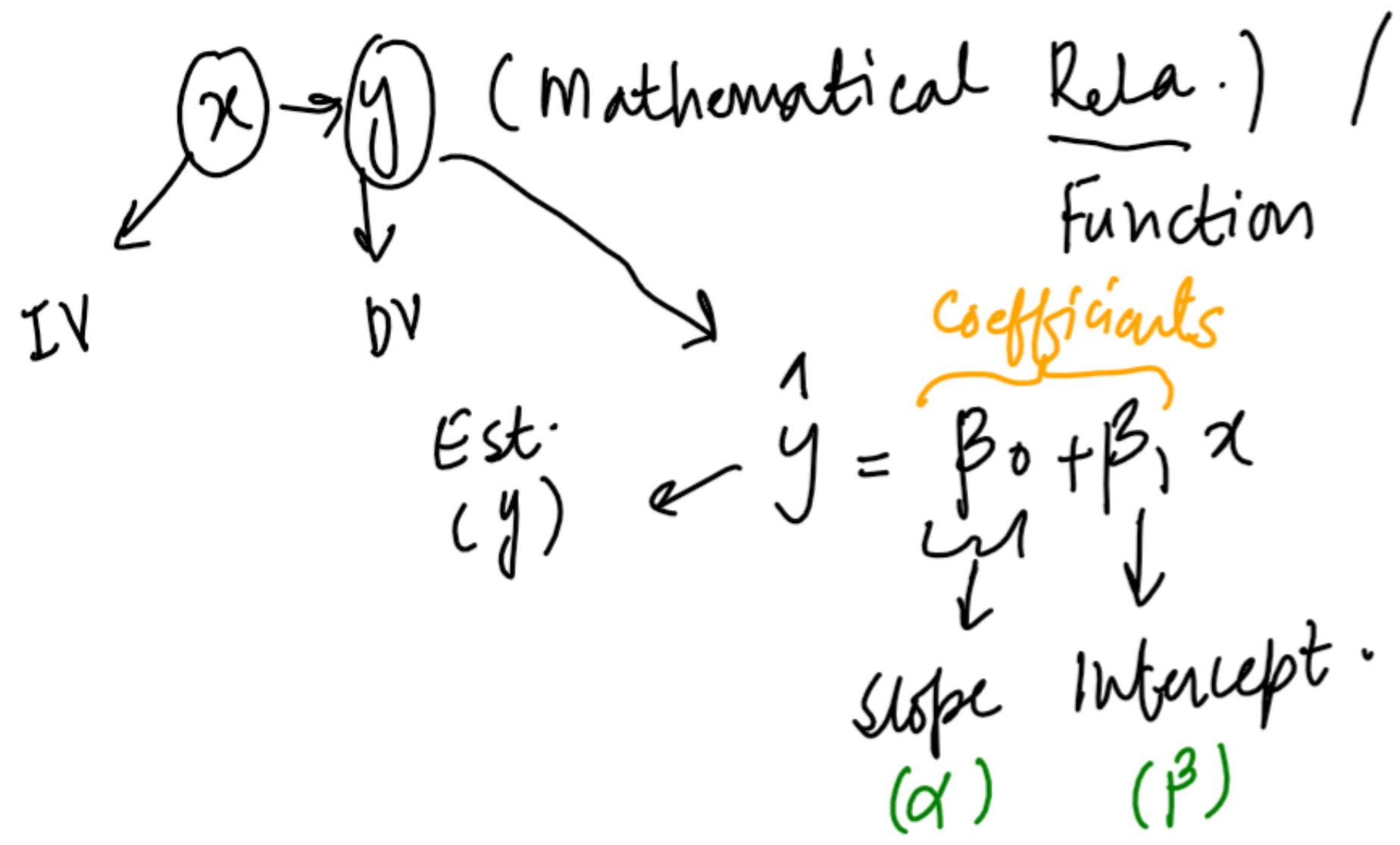
random error.

Sum of Square of Error
SSE = $\sum (\hat{y} - y)^2$

x	y	$\hat{y} = -6.2 + 2.6(x)$
5	8	$\hat{y} = -6.2 + 2.6(5) = 6.8$
6	9	$\hat{y} = -6.2 + 2.6(6) = 9.4$
3	2	$\hat{y} = -6.2 + 2.6(3) = 1.6$
4	3	$\hat{y} = -6.2 + 2.6(4) = 4.2$

SIMPLE LINEAR REGRESSION: (Part-II)

$$\bar{x} = \frac{\sum x}{n}; \bar{y} = \frac{\sum y}{n}$$



$$\beta_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n x_i^2 - \frac{(\sum_{i=1}^n x_i)^2}{n}}$$

$$\beta_0 = \frac{1}{n} \left(\sum_{i=1}^n y_i - \beta_1 \sum_{i=1}^n x_i \right) = \bar{y} - \beta_1 \bar{x}$$

$$= \left(\frac{\sum y_i}{n} \right) - \beta_1 \left(\frac{\sum x_i}{n} \right)$$

$\bar{y}$ $\bar{x}$

- Mathematical function in modeling / forecasting.
- Unknown / new variable info

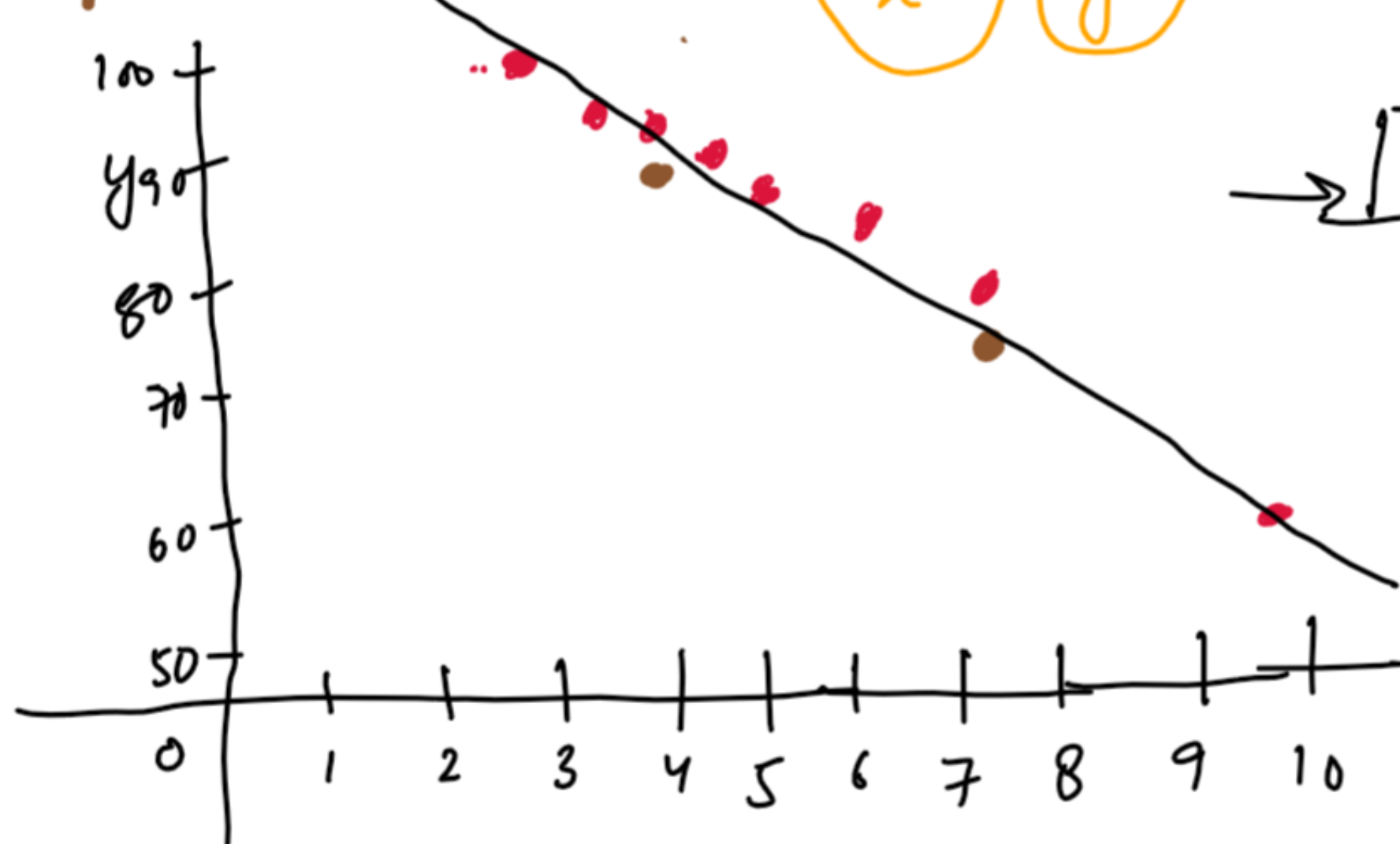
PHENOMENA MUST BE LINEAR

Exp: 1

14.98 Tax Efficiency. Following are the data on percentage of investments in energy securities and tax efficiency from Exercise 14.58.

Invest.	3.1	3.2	3.7	4.3	4.0	5.5	6.7	7.4	7.4	10.6
eff. ying.	98.1	94.7	92.0	89.8	87.5	85.0	82.0	77.8	72.1	53.5

DV / OUTCOME



PRIOR INFORMATION

LINEARITY

- 100
- 116 $\rightarrow$ 18% GST.
- 98
- 97
- 60
- 170

14.98 Tax Efficiency. Following are the data on percentage of investments in energy securities and tax efficiency from Exercise 14.58.

x	3.1	3.2	3.7	4.3	4.0	5.5	6.7	7.4	7.4	10.6
y	98.1	94.7	92.0	89.8	87.5	85.0	82.0	77.8	72.1	53.5

n = 10

x	y
3.1	98.1
3.2	94.7
3.7	92.0
4.3	89.8
4.0	87.5
5.5	85.0
6.7	
7.4	
7.4	
10.6	

$$\sum x = 55.9, \sum y = 832.5$$

$$\sum x = 55.9$$

$$\sum y = 832.5$$

$$\sum xy = 4476.95$$

$$\sum x^2 = 365.05$$

$$\hat{y} = \beta_0 + \beta_1 x$$

$$= 102.03 + (-3.36)x$$

$$\hat{y} = 102.03 - 3.36x$$

$$\beta_1 = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum x_i^2 - \frac{(\sum x_i)^2}{n}}$$

$$= \frac{\sum x_i y_i - \frac{\sum x_i \sum y_i}{n}}{\sum x_i^2 - \frac{(\sum x_i)^2}{n}}$$

$$\beta_0 = \frac{1}{n} \left(\sum_{i=1}^n y_i - \beta_1 \sum_{i=1}^n x_i \right) = \bar{y} - \beta_1 \bar{x}$$

$$\beta_0 = \frac{\sum y_i}{n} - \beta_1 \frac{\sum x_i}{n}$$

$$\beta_1 = \frac{4476.95 - \frac{(55.9)(832.5)}{10}}{(365.05) - 312.48}$$

$$\beta_1 = \frac{4476.95 - 4653.67}{(365.05) - (312.48)} = \frac{-176.72}{52.57}$$

$$\beta_1 = -3.36$$

$$\beta_0 = 83.25 - (-3.36)(5.59)$$

$$= 83.25 + 18.78$$

$$= 102.03$$

$$(55.9)^2 = 3124.81$$

14.98 Tax Efficiency. Following are the data on percentage of investments in energy securities and tax efficiency from Exercise 14.58.

x	3.1	3.2	3.7	4.3	4.0	5.5	6.7	7.4	7.4	10.6
y	98.1	94.7	92.0	89.8	87.5	85.0	82.0	77.8	72.1	53.5

x	y	$\hat{y} = 102.03 - 3.36x$	$e = y - \hat{y}$	$(e)^2$
3.1	98.1	$\hat{y}_1 = 102.03 - 3.36(3.1) = 91.61$	$e_1 =$	$(e_1)^2$
3.2	94.7	$\hat{y}_2 = 102.03 - 3.36(3.2) = 91.37$	$e_2 =$	
3.7	92.0			
4.3	89.8			
4.0	87.5			
5.5	85.0			
6.7	82.0			
7.4	77.8			
7.4	72.1			
10.6	53.5			

y = Original Value
 $\hat{y}$ = Estimated Values

Sum of Error = 0

Error
 $e_i = y - \hat{y}$
 $e_i^2 = (y - \hat{y})^2$

SSE: sum of square of error

$$SSE = \sum (y_i - \hat{y}_i)^2$$

Loss

$$\hat{y} = \beta_0 + \beta_1 x$$

$$\sum (y_i - \hat{y}_i) = 0$$

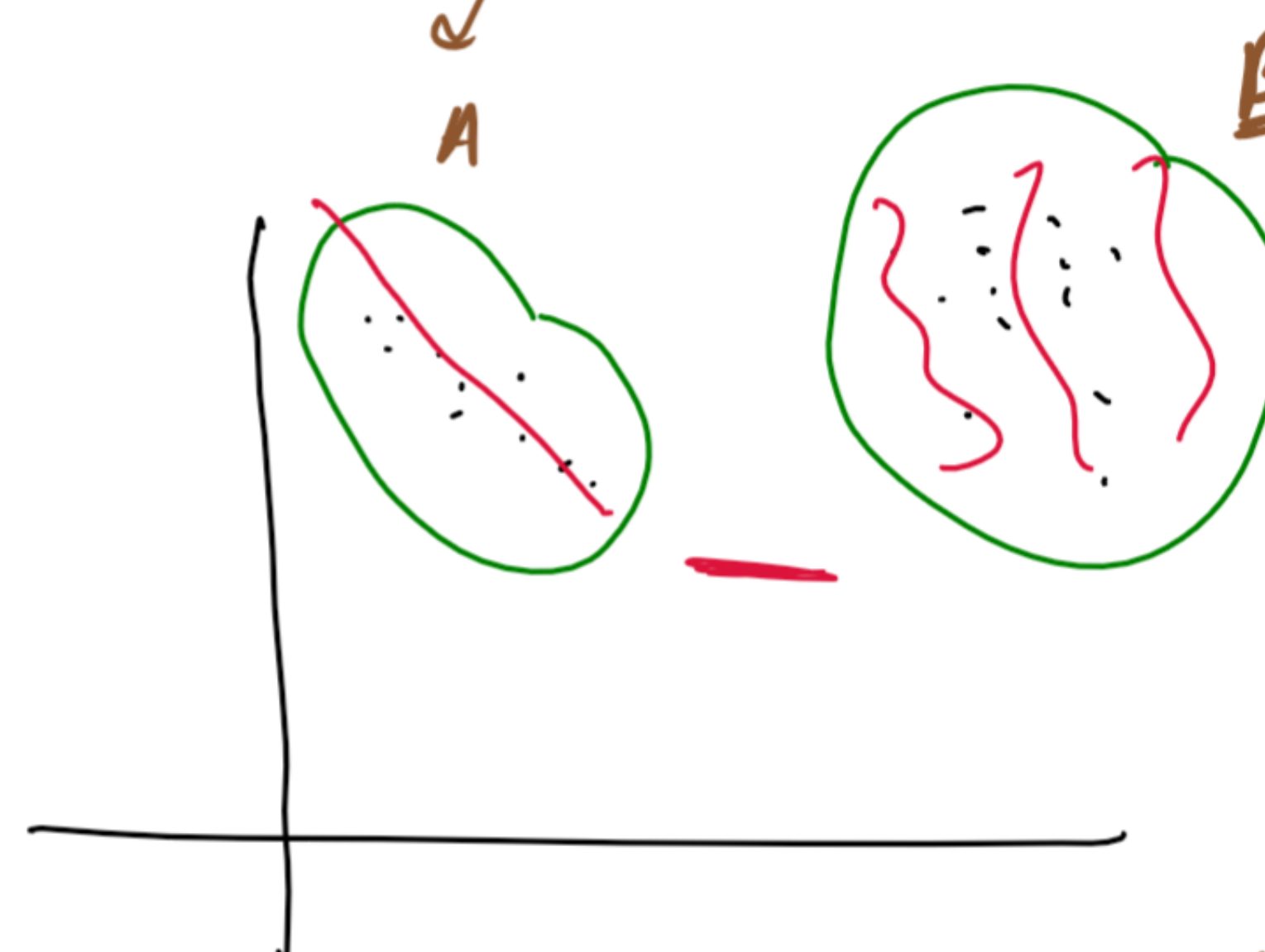
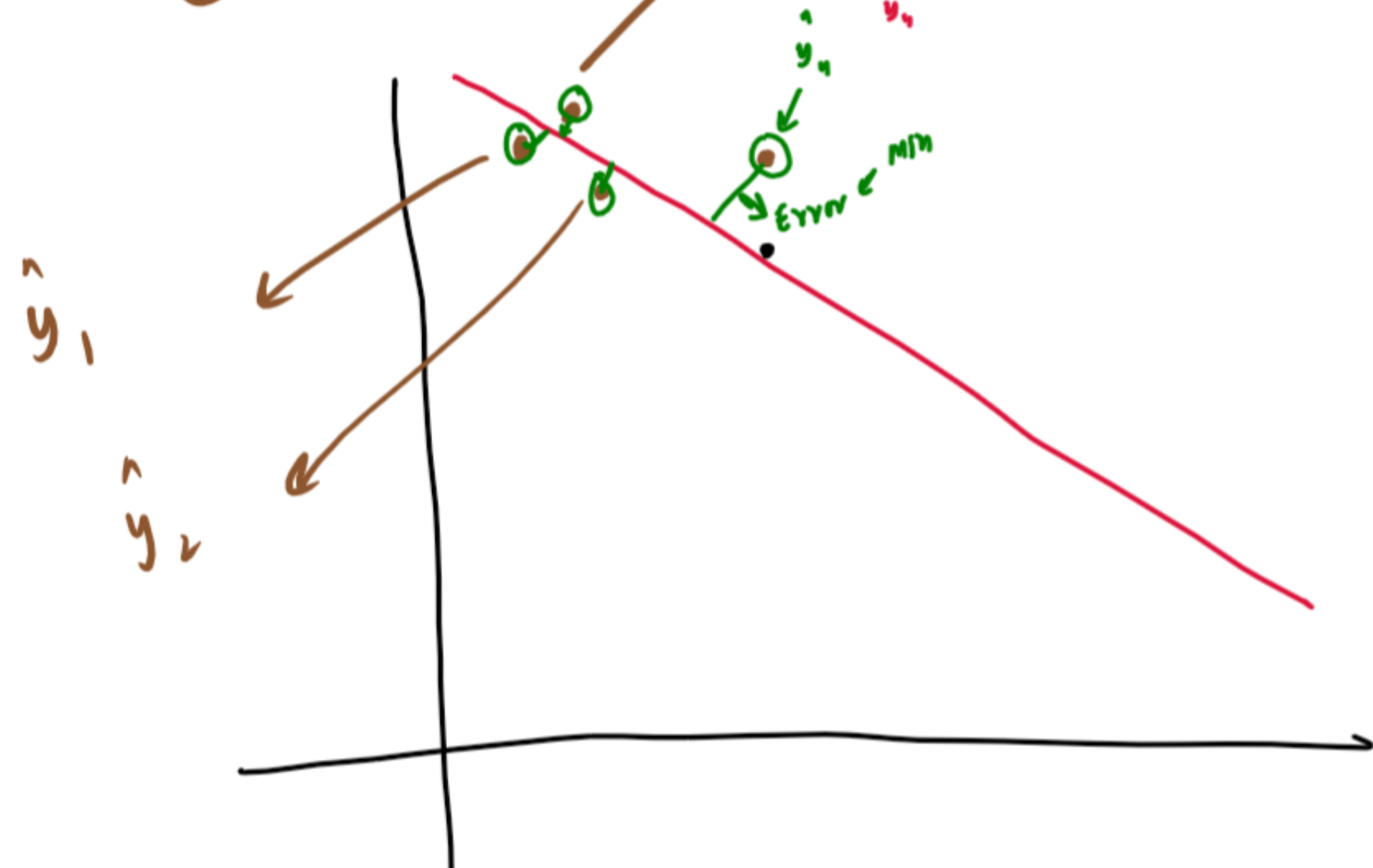
$$\sum (y_i - \hat{y}_i)^2$$

$$Loss = \sum (y_i - \beta_0 - \beta_1 x)^2$$

LR

Non-Linear

$$\hat{y} = \beta_0 + \beta_1 x + e_i \quad \hat{y}_3$$



Logistic



COEFFICIENT OF DETERMINATION (r^2) (STRENGTH OF MODEL).

Tells how much variation in y is explained by x .

2) Total sum of Squares = $TSS = \sum_{i=1}^n (y_i - \bar{y})^2$

3) Regression sum of Squares = $SSR = \sum_{i=1}^n (\hat{y}_i - \bar{y})^2$

$\bar{y} = \frac{\sum y}{n}$	① x	② y_i	③ $\hat{y}_i = \beta_0 + \beta_1 x$	④ $(y_i - \bar{y})^2$	⑤ $(\hat{y}_i - \bar{y})^2$
				$\downarrow$	$\downarrow$
	$\sum x$	$\sum y$		$\sum (y_i - \bar{y})^2$ TSS	$\sum (\hat{y}_i - \bar{y})^2$ SSR

y_i = actual/original
 $\bar{y}$ = Mean of y or Average.
 $\hat{y}$ = Estimated y (from model)

① = $SS E = \sum_{i=1}^n (y_i - \hat{y}_i)^2$

Correlation
 $-1 \leq r \leq 1$

Co Det. $\rightarrow r^2 = \frac{SSR}{TSS} = \frac{\sum_{i=1}^n (\hat{y}_i - \bar{y})^2}{\sum_{i=1}^n (y_i - \bar{y})^2}$

Square of correlation

$\rightarrow 0 \leq r^2 \leq 1$

$r^2 \sim 0$ (Model is not good for prediction)
 $r^2 \sim 1$ (Model is good for prediction)

Model ✓
 Validate -
 Evaluate -

eg $r^2 = 0.85 \rightarrow 85\%$

TRAINING AND TESTING DATA.

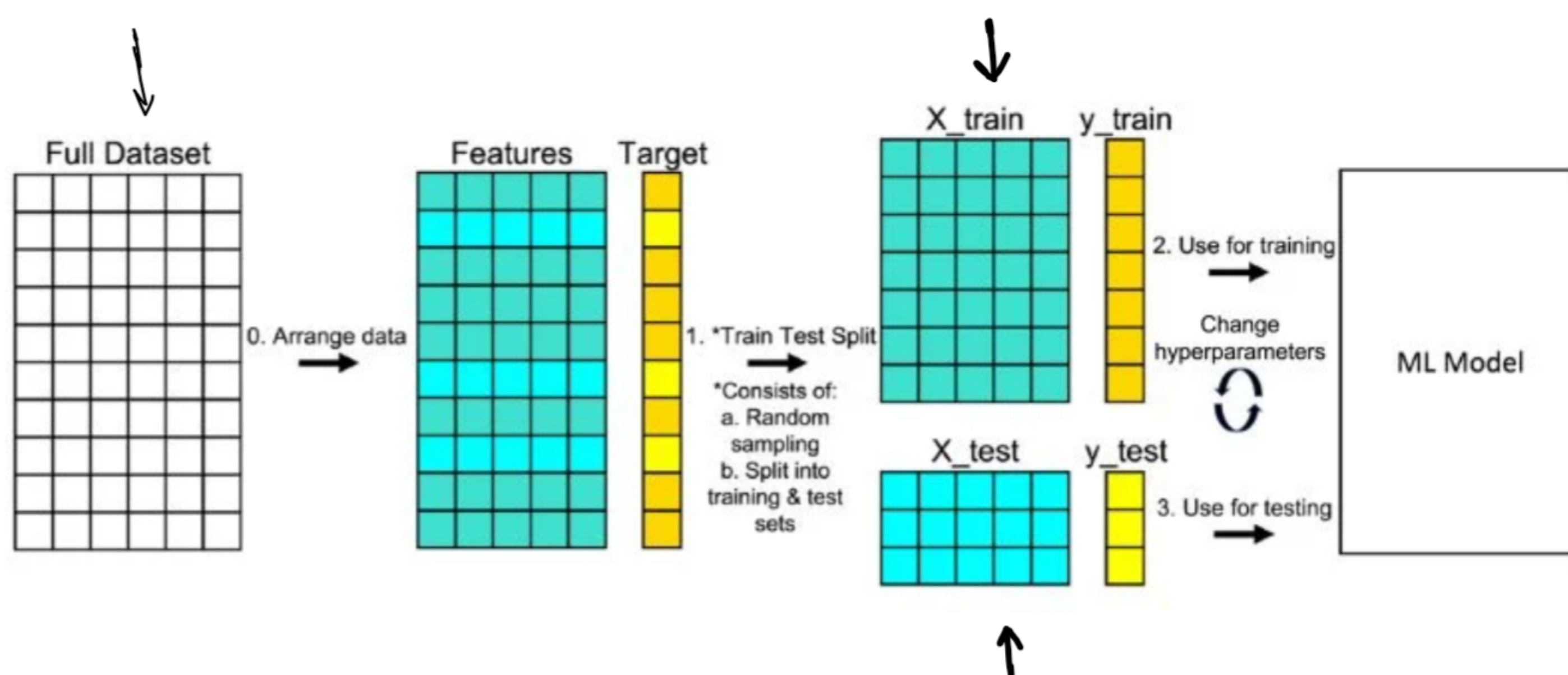
- Minimize expected prediction error.
- Basic data splitting rule

↓ MOST COMMON

small (< 1000) → 70% Train, 30% Test

medium → 80% Train, 20% Test

large → 90% Train, 10% Test



EVALUATION METRICS FOR LINEAR REGRESSION

- MEAN SQUARE ERROR (MSE)

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

x	y	$\hat{y}$	$(y - \hat{y})^2$
1	56		
2	71		
3	60		
4	75		
5	77		
6	80		
7	83		
8	87		

- MEAN ABSOLUTE ERROR (MAE)

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

x	y	$\hat{y}$	$ y_i - \hat{y} $
1	56		
2	71		
3	60		

- ROOT MEAN SQUARED ERROR (RMSE)

$$RMSE = \sqrt{\frac{SSR}{n}}$$

$$SSR = \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

- COEFFICIENT OF DETERMINATION

$$R^2 = 1 - \left(\frac{SSR}{SST} \right)$$

$$SST = \sum_{i=1}^n (y_i - \bar{y})^2$$

ADJUSTED R^2 ERROR

$$Adj R^2 = 1 - \left[\frac{(1 - R^2) \cdot (n - 1)}{n - K - 1} \right]$$

K = no. of predictors in the model.
"features"

Example of Linear Regression:

Train: $\beta_1 = \frac{\sum xy - \frac{\sum x \sum y}{n}}{\sum x^2 - \frac{(\sum x)^2}{n}}$

Test: $\hat{y} = 61 + 0.2x$

Train Error: $e = y - \hat{y}$

Testing Error: $e = y - \hat{y}$

Train Error $\approx$ Train Error → Good fit

Train Error low

Train Error high $\approx$ overfit

Train Error high $\approx$ underfit

Train "low"

$$\hat{y} = \beta_0 + \beta_1 x + \epsilon \quad K=1$$

$$\hat{y} = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \epsilon \quad K=2$$

size	No. of beds	y = Price
x_1	x_2	
15	3	150
16	2	700
18	4	200

MULTIPLE LINEAR REGRESSION

Target Var. $\downarrow$ Est. Var. $\uparrow$
 $error = y_i - \hat{y}$

$$\hat{y} = \beta_0 + \beta_1 \underline{x_1} + \beta_2 \underline{x_2} + \epsilon$$

EXAMPLE:

- Model / predict house price (y)

using:

- x_1 = size of house (in 1000 sq. ft.)
- x_2 = no. of bedrooms.

$$\hat{y} = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \epsilon$$

$$\hat{y} = \underline{X} \underline{\beta} + \epsilon$$

$$\hat{\text{House price}} = \beta_0 + \beta_1 (\text{size of house}) + \beta_2 (\text{no. of bedrooms}) + \epsilon$$

y PRICE	(x_1) SIZE OF HOUSE	(x_2) NO. OF BEDROOMS
250	1.5	2
350	2	3
650	4	6

- Convert it to matrix form

$$X = \begin{bmatrix} 1 & 1.5 & 2 \\ 1 & 2 & 3 \\ 1 & 4 & 6 \end{bmatrix} ; Y = \begin{bmatrix} 250 \\ 350 \\ 650 \end{bmatrix}$$

$\downarrow$
OUT

$\underline{10 \times 3}$ $\underline{3 \times 3}$ $\underline{3 \times 1}$

DUE TO MATRIX FORM

$$\hat{y} = X \hat{\beta}$$

DUE TO MATRIX FORM
ORDINARY LEAST SQUARE (OLS)
 METHOD WILL BE USED.

Ordinary Least Square (OLS):

$$\hat{y} = X \hat{\beta}$$

OLS $\rightarrow \hat{\beta} = (X^T X)^{-1} X^T y$

$$X = \begin{bmatrix} 1 & 1.5 & 2 \\ 1 & 2 & 3 \\ 1 & 4 & 6 \end{bmatrix}$$

$$X^T X = \begin{bmatrix} 1 & 1 & 1 \\ 1.5 & 2 & 4 \\ 2 & 3 & 6 \end{bmatrix} \cdot \begin{bmatrix} 1 & 1.5 & 2 \\ 1 & 2 & 3 \\ 1 & 4 & 6 \end{bmatrix}$$

$$= \begin{bmatrix} 3 & 7.5 & 11 \\ 7.5 & 23 & 32.5 \\ 11 & 32.5 & 49 \end{bmatrix}$$

TASK TO FIND THIS SOLUTION

FROM THE SOLUTION OF THIS WE WILL HAVE

$$\hat{\beta} = \begin{bmatrix} \hat{\beta}_0 \\ \hat{\beta}_1 \\ \hat{\beta}_2 \end{bmatrix}$$

$\rightarrow$ This will form the equation as

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x + \hat{\beta}_2 x + \epsilon$$

$$\therefore \text{here } \epsilon = y - \hat{y} = \begin{bmatrix} \end{bmatrix}_{3 \times 1} - \begin{bmatrix} \end{bmatrix}_{3 \times 1}$$

$$SSE = \sum_{i=1}^n e_i^2$$

$$SST = \sum (y_i - \bar{y})^2$$

$$\text{And } R^2 = 1 - \frac{SSE}{SST}$$

$$\bar{y} = \frac{\sum x}{n}$$

For 2x2

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

$$A^T = \begin{bmatrix} a & c \\ b & d \end{bmatrix}$$

$$(A^T)^{-1} = \frac{1}{|A^T|} \text{Adj}(A^T)$$

$$|A^T| = ad - bc$$

$$\text{adj}(A^T) = \begin{bmatrix} d & -c \\ -b & a \end{bmatrix}$$

$$\hat{y} = X \hat{\beta}$$

$$X = \begin{bmatrix} x_0 \\ x_1 \\ x_2 \end{bmatrix}$$